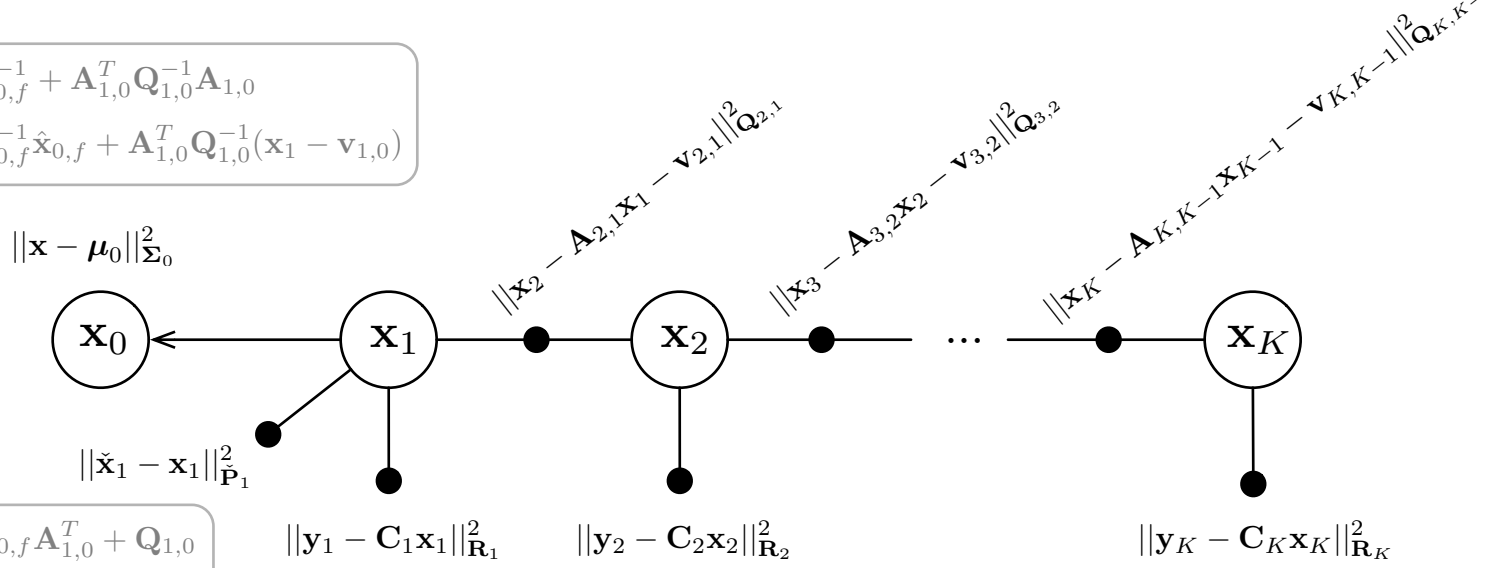
$$\boldsymbol{\Sigma}_0^{-1} = \hat{\mathbf{P}}_{0,f}^{-1} + \mathbf{A}_{1,0}^T \mathbf{Q}_{1,0}^{-1} \mathbf{A}_{1,0}$$

$$\boldsymbol{\Sigma}_0^{-1} \boldsymbol{\mu}_0 = \hat{\mathbf{P}}_{0,f}^{-1} \hat{\mathbf{x}}_{0,f} + \mathbf{A}_{1,0}^T \mathbf{Q}_{1,0}^{-1} (\mathbf{x}_1 - \mathbf{v}_{1,0})$$



$$\check{\mathbf{P}}_1 = \mathbf{A}_{1,0} \hat{\mathbf{P}}_{0,f} \mathbf{A}_{1,0}^T + \mathbf{Q}_{1,0}$$

$$\check{\mathbf{x}}_1 = \mathbf{A}_{1,0} \hat{\mathbf{x}}_{0,f} + \mathbf{v}_{1,0}$$